FINAL GROUP ASSIGNMENT (35%)

Due: 17 JANUARY 2025, Friday, 11:59 PM (GMT +7)

INTRODUCTION

In this group assignment, you will collaboratively design and implement a credit-based auction application using C++, focusing on developing key functionalities outlined in the Description section.

OBJECTIVES

This assignment offers students the opportunity to apply the concepts learned throughout the course so far, providing hands-on experience in analyzing, designing, and developing a software application. The project aims to simulate real-world development scenarios, helping you prepare for professional life. For simplicity, the application will use a basic text interface (no GUI required), but it will include all essential functionalities of a practical application, with potential for further development after completion.

DESCRIPTION

AUCTION APPLICATION

Develop a console-based auction application in C++ that allows users to buy, sell and bid on items using a Credit Point (CP) system. The application should provide functionalities for member registration, profile management, item listings, placing bids, and ratings while ensuring security and scalability.

Member registration: Non-members (as guests) must register as members by providing their personal information: username, password, full name, phone number, email, ID type (Citizen ID/Passport) and ID/Passport number. To ensure account security, you must implement a strong password policy. Simple passwords like “12345” or “password” are weak password and will not be accepted. Registration is free of charge and the initial CP balance for new members will be set to 0.

Profile management: Members can view and update their profile information, including the ability to change their password and check their current CP balance. Members can top up their credit points at a rate of $1 per credit point.

Item listings: Members can act as both buyers and sellers. Sellers can create item listings by providing the following details: item name, category, short description, starting bid, bid increment (the minimum amount by which bids must increase), end date & time of the auction and minimum buyer rating score (refer to Ratings and Reviews section). The application will automatically record the start date & time when the seller creates the listing. All transactions will be conducted using credit point (CP). Sellers can edit/remove their listings only if no active bids have been placed yet. Once bids have started, listings cannot be modified or removed.

Bidding listed item: Buyers can perform the following actions:

- Search for items by name, category, or credit point range to find relevant listings.
- View item details, including the current highest bid, the highest bidder, and the auction’s end date & time.
- Place a bid if they have enough credit points. If a buyer has multiple active bids, the application should ensure that they have sufficient credit points to cover all their ongoing bids before allowing any new bids.

Auction conclusion: The highest bid at the end of the auction wins. Credit points are deducted from the winner's account and transferred to the seller's account upon conclusion of the auction.

Ratings and Reviews: The application implements a rating system with scores ranging from 1 to 5 for both buyers and sellers.

- **After a transaction:** Buyers rate sellers based on item quality and communication during/after the transaction (assuming there was communication between the buyer and seller). Sellers rate buyers based on communication. The final rating for each party is the average of all ratings they receive.
- **New members:** Start with a default rating of 3 for both their seller and buyer roles.
- **Impact of ratings on listings:**
  (1) Sellers can restrict buyers with low ratings from viewing their item listings by setting a minimum rating threshold when creating the listing.
  (2) Buyers can view seller ratings on item listings to help them decide whether to place a bid.
  (3) The search results will only display items where the buyer meets the minimum rating threshold set by the seller, ensuring that buyers can only see listings that match their rating profile.

Cancellation: Buyers cannot cancel or withdraw their bid once they have placed a bid on an item. Likewise, sellers cannot cancel or remove their item listings once a bid has been placed on the item.

Non-members: Non-members (as guests) can only view limited information on item listings: item name, category, and a short description. To access additional features and fully benefit from the application, guests must register as members.

Administrator privilege: Admin can view all member profiles and item listings.

CONSTRAINTS:

(1) Credit point balances must never go negative.
(2) Minimum number of item categories is 3, ensuring a diverse range of listings.

ASSUMPTIONS:

(1) Transactions involving the top-up of credit points with cash can only be authorized by the member's password for security purposes.
(2) New members start with a default rating of 3. To increase their ratings, members are encouraged to engage in buying and selling items.

PROCESS AND GUIDE

A. Class Diagram: Create a class diagram that identifies the classes, attributes, methods, and the relationships between the classes for the auction application.

B. *Basic Features:*

Implement and test the following core features for the auction application. **Ensure that all user input is validated appropriately. All attributes must be private to enforce data encapsulation.**

1. Non-member (as guest) can view item listings with limited details: item name, category, and short description.

2. Non-member (as guest) can register to become a member. Upon registration, they will start with 0 credit points and a default rating of 3 (all information is recorded and stored). The registration process must include a strong password policy.
3. A member can log in with registered username and password. They can view and update their profile information, including changing password and topping-up credit points.
4. Admin can view all member profiles and item listings without restrictions.
5. A member (as seller) can create, edit, or remove item listings only if no active bids have been placed on those items.
6. A member (as buyer) cannot cancel their bid once it has been placed on an item.
7. A member (as buyer) can search items by name, category, or credit point range. The search should filter items according to their rating score. For example, if a buyer's rating score is 2 and an item requires a minimum rating score of 3, that item will not appear in the search results.
8. A member (as buyer) can view item details including the current highest bid, the current bidder and the auction's end date & time.
9. A member (as buyer) can place bids on items, provided they have sufficient credit points.
10. A member (as buyer) can win the auction if they place the highest bid. When the auction concludes, the credit points of both the buyer and seller should be updated accordingly.
11. A member can perform the ratings and reviews after completing a transaction.
12. A member can view their own average rating, as well as reviews left by others.
13. All data must be saved into **data file(s)** before the application ends. Upon starting the application, this data should be loaded and made available for use.

C. ***Advanced Features:***

14. ***Activity dashboard:*** When a member logs into the application, a personalized dashboard should be displayed showing their account overview, active item listings and active bids. Below is a sample dashboard. You can design your own dashboard format as needed.

```
Account Overview: iAmMember1
----------------------------
Active Item Listings: 3   |  Active Bids: 2
Items Won to Date: 1      |  Account Balance: 300 CP

Your active item listings
-------------------------
No. | Name           | Category     | Current bid  | Current bidder     | End Date & Time
1   | iPhone 13 Pro  | Electronics  | 350          | iAmMember12        | 16/12/2024 18:00
2   | Rare coin set  | Collectibles | 300          | iAmMember4         | 20/12/2024 09:00
3   | PlayStation 4  | Games        | No bids      | No bidders         | 25/12/2024 08:00

Your active bids
----------------
No. | Name           | Category     | Your bid     | Current bid        | End Date & Time
1   | Sony headphones| Electronics  | 100          | 250                | 20/12/2024 10:00
2   | Road bike      | Vehicles     | 300          | 310                | 23/12/2024 17:30
```

15. ***Automatic bid limit***:

(a) **Bid limit functionality:** Buyers can set an automatic bid limit for an item they are bidding on. The application will automatically increment their bid until their set limit is reached. For example, let's say an item A has a bid increment of 10 CP.

Scenario (1):

i. Buyer XXX placed an initial bid of 50 and set a bid limit of 90.
ii. Buyer YYY placed a bid of 60. The application automatically places a bid of 70 for buyer XXX.

iii. Buyer ZZZ placed a bid of 80. The application automatically places a bid of 90 for buyer XXX again.
iv. Buyer WWW placed a bid of 100. Since buyer XXX's limit is 90, no further automatic bid is placed.

(b) **Automatic bid limit conflict:** Automatic bid limits cannot be identical for multiple buyers on the same auction item. If a new buyer sets an automatic bid limit equal to an existing bid limit, the application will reject the limit and prompt them to adjust the limit to the next available bid increment. Scenario (2):

i. Buyer XXX sets an automatic bid limit of 90.
ii. Buyer WWW tries to set the same limit of 90.
iii. The application rejects buyer WWW's limit and prompts to change it to 100, as the bid increment is 10.

D. Welcome Screen

When starting the application, display a welcome screen with an example content structure as shown below. You may adjust the format and content as needed. **Note:** the **bold** content is compulsory, and **"…"** indicates that you can add additional information as needed.

```
EEET2482/COSC2082 GROUP ASSIGNMENT
Semester 3 2024
AUCTION APPLICATION

Instructor: Dr Ling Huo Chong
Group: Group No.
XXXXXXX, Student Name
XXXXXXX, Student Name
XXXXXXX, Student Name
XXXXXXX, Student Name

Use the app as 1. Guest   2. Member   3. Admin

Enter your choice: 2
Enter username: iAmMember1
Enter password: XXXXXXXXXX

You have successfully logged in.

Account Overview: iAmMember1
----------------------------
Active Item Listings: 3  |  Active Bids: 2
Items Won: 1             |  Account Balance: 300 CP

Your active item listings
-------------------------
No. | Name           | Category     | Current bid  | Current bidder     | End Date & Time
1   | iPhone 13 Pro  | Electronics  | 350          | iAmMember12        | 16/12/2024 18:00
2   | Rare coin set  | Collectibles | 300          | iAmMember4         | 20/12/2024 09:00
3   | PlayStation 4  | Games        | No bids      | No bidders         | 25/12/2024 08:00

Your active bids
----------------
No. | Name           | Category     | Your bid     | Current bid        | End Date & Time
1   | Sony headphones| Electronics  | 100          | 250                | 20/12/2024 10:00
2   | Road bike      | Vehicles     | 300          | 310                | 23/12/2024 17:30

This is your menu:
0.  Exit
1.  View Information
2.  ...
Enter your choice: 1
...
```

IMPORTANT INSTRUCTIONS

1. You are required to write your program in C++ programming language only.
2. You will need to demonstrate the OOP skills that you have learned on the course.
3. You must thoroughly document your code using comments (// or /* … */).
4. Your code should be separated logically into multiple source code (.cpp) and header (.h) files that overall have a single purpose, i.e., your code should be structured in a way that each file has a clear scope and goal. Marks will be deducted if you put everything together into one single source code file. Make sure you use the header files and header guard correctly.

USING GITHUB

You are recommended to use GitHub for your group assignment. GitHub is a widely used platform for version control and project collaboration. It allows team members to collaborate effectively, manage code efficiently, and track issues seamlessly. For guidance on using GitHub, please refer to the following tutorial: https://github.com/TriHuynh00/github-demo.

ACADEMIC INTEGRITY

This assessment requires that you meet RMIT's expectations for academic integrity. You must not ask for or accept help from anyone else on completing the tasks. You must not show your work to another student/group enrolled in this course who might gain unfair advantage from it. These things are considered plagiarism or collusion. Plagiarism is a form of cheating. It is the presentation of the work, idea, or creation of another person as though it is your own. If you use any resources, write an acknowledgment in the file where the resources were used. Any submission found in whole or in part plagiarized will be considered as plagiarism. To be on the safe side, never ever release your code to the public. For example, **if you use GitHub to store your code, make sure that you set the repository as private.** For more information, visit RMIT's website for Academic Integrity.

The penalty for plagiarism is extremely severe at RMIT. If you are found to plagiarize, you will receive a mark of zero for an assessment or even an entire course. Repeated plagiarism will lead to exclusion from RMIT.

RUBRIC

Refer to the Rubric published in the Group Assignment page in the course Canvas.

REPORT AND PRESENTATION

Report: Complete your report with a class diagram and detailed explanation using the provided Report Template available on the Group Assignment page in the course Canvas.

Video Presentation: Record a video, around 10 minutes and less than 15 minutes, showcasing and demonstrating your work. Ensure that all members participate in the presentation video. The video presentation should introduce the topic and its significance, and then demo your program with explanations/assumptions that you have made in your program. You must upload your video to YouTube.

SUBMISSION INSTRUCTIONS

As a group, submit the following files separately to the course Canvas, DON'T zip them together (*only ONE member needs to do the submission***):**

1. An executable file of your program, i.e., ***GroupX_Program.exe*** *(where X denotes your group number).*
2. A zip file of all source code including .h and .cpp files and data files, i.e., ***GroupX_Sources.zip.***
3. A report in PDF format, i.e., ***GroupX_Report.pdf***.
4. A text file with a link to your presentation video, i.e., ***GroupX_Video_GitHub.txt.*** If your group is using GitHub, put the link of your GitHub to the text file as well, and invite your lecturer to your GitHub, i.e., **louislinghc** (username for Dr Ling).
5. Record of Individual Contribution in PDF format, i.e., ***GroupX_Individual_Contribution.pdf.*** Refer to the Individual Contribution form available on the Group Assignment page in the course Canvas. **Note:** The commit history on GitHub can serve as proof of members' participation in the group assignment if there is a dispute.

Note: *The files must also be submitted separately to TURNITIN, and make sure your executable file can run. If you do not submit it, or it fails to run/crashes, you may lose up to 50% of the grade (although some marks may be given for the effort).*

Late work: 10% penalty per day. Submission later than 03 days will not be accepted and a mark of zero will be given.

--- END OF ASSIGNMENT ---